1 (a)	Attempts $u_3 = 3u_2 - 2u_1 \Rightarrow 4 = 3u_2 - 2\times 7 \Rightarrow u_2 = ...$	M1
	$\Rightarrow u_2 = 6$	A1
		(2)
(b)	Attempts $u_4 = 3u_3 - 2u_2 = 3\times 4 - 2\times "6" = (0)$	M1
	$\sum_{r=1}^{4}(u_r + 2r) = (7 + 2\times 1) + ("6" + 2\times 2) + (4 + 2\times 3) + ("0" + 2\times 4)$	dM1
	$= 37$	A1
		(3)
		Total 5

2 (a)	Strip width = 1.5	B1
	$\frac{3}{4}\{4.16+2.28+2\times(2.91+a+1.73+1.37+1.43)\}=19.3\Rightarrow a=...$	M1
	a = awrt 2.21	A1
		(3)
(b)	$\int_{-4}^{5}(2\,f(x)-3)\,dx=2\times19.3-\left[3x\right]_{-4}^{5}$	M1
	$=11.6$	A1
		(2)
		Total 5

3(a)	Attempts to complete the square for both variables $\left(x+4\right)^2, \left(y-7\right)^2$	M1
	Centre $\left(-4, 7\right)$	A1
	Radius = 12	A1
		(3)
(b)	Attempts $"\pm\left(12"-\sqrt{"4"^2+"7"^2}\right)$	M1
	$12-\sqrt{65}$	A1 ft
		(2)
		Total 5

4 (a)	$(3+2x)^6$	
	First term 3^6 or 729	B1
	Term in x, x^2 or x^3: Award for one of ${}^6C_5(3)^5(2x)^1$, ${}^6C_4(3)^4(2x)^2$ or ${}^6C_3(3)^3(2x)^3$	M1
	Two of $\ldots+2916x+4860x^2+4320x^3+\ldots$	A1
	$(3+2x)^6=729+2916x+4860x^2+4320x^3+\ldots$	A1
		(4)
(b)	Attempts one correct term $2x^2\times"729"$ **or** $\pm\frac{1}{6x}\times"4320"x^3$	M1
	Attempts to combine the correct two terms $2x^2\times"729"\pm\frac{1}{6x}\times"4320"x^3=\ldots x^2$	dM1
	738 but condone $738x^2$	A1
		(3)
		Total 7

5 (a)	$D = 8 + 5\sin\left(\frac{\pi \times 2}{6} + 3\right) = 4.07$	B1
		(1)
(b)	(b) $6 = 8 + 5\sin\left(\frac{\pi t}{6} + 3\right) \Rightarrow \sin\left(\frac{\pi t}{6} + 3\right) = -\frac{2}{5}$	M1 , A1
	$\Rightarrow \left(\frac{\pi t}{6} + 3\right) = \arcsin\left(-\frac{2}{5}\right) =$ Any of $3.55,\ 5.87, 9.84, 12.2$	dM1
	$\Rightarrow t =$ Any of $\frac{6(3.55-3)}{\pi}, \frac{6(5.87-3)}{\pi}, \frac{6(9.84-3)}{\pi}, \frac{6(12.2-3)}{\pi}$	ddM1
	13:04 or 1:04 pm	A1
		(5)
		Total 6

6(a)	Sets $\mathrm{f}\left(-\frac{3}{2}\right)=0 \Rightarrow \left(9p+4q=102\right)$	M1
	Sets $\mathrm{f}(-2)=-5 \Rightarrow \left(4p+q=43\right)$	M1
	Solves to get values for p and q	dM1
	(i) $p=10$ * (ii) $q=3$ following two correct equations	A1*, A1
		(5)
(b)	$\mathrm{f}'(x)=12x^2+20x+8$	B1
	Solves $\mathrm{f}'(x)=0 \Rightarrow 4\left(3x+2\right)(x+1)=0 \Rightarrow x=-\frac{2}{3},-1$	M1, A1
	$-1<x<-\frac{2}{3}$	A1
		(4)
		Total 9

7 (i)	States or uses $\tan x = \dfrac{\sin x}{\cos x}$	M1
	$3\sin x \tan x = 11 + \cos x \Rightarrow 3\sin^2 x = 11\cos x + \cos^2 x$	
	$\Rightarrow 3\left(1-\cos^2 x\right) = 11\cos x + \cos^2 x \Rightarrow 4\cos^2 x + 11\cos x - 3 = 0$	dM1, A1
	$\Rightarrow \left(4\cos x - 1\right)\left(\cos x + 3\right) = 0 \Rightarrow \cos x = \dfrac{1}{4} \Rightarrow x = 1.318, 4.965$	dM1 A1
		(5)
(ii)	$\cos\theta = \dfrac{1}{3} \Rightarrow \sin^2\theta = 1 - \dfrac{1}{3^2} \Rightarrow \tan\theta = \dfrac{\sin\theta}{\cos\theta} = \dfrac{\sqrt{1-\dfrac{1}{3^2}}}{\dfrac{1}{3}}$	M1
	$\Rightarrow \tan\theta = 2\sqrt{2}$	A1
		(2)
		Total 7

8 (i)	(a) $S_n = a+(a+d)+.....+(a+(n-1)d)$ (1) $S_n = (a+(n-1)d)+.....(a+d)+a$ (2) (1) + (2) $2S_n = (2a+(n-1)d)+(2a+(n-1)d).....+(2a+(n-1)d)$ $2S_n = n(2a+(n-1)d) \Rightarrow S_n = \frac{n}{2}(2a+(n-1)d)$ *	B1 M1 A1*
		(3)
	(b) States or uses two of $a = 900$, $d = -8$ and $n = 51$ Or alternatively states or uses two of $a = 500$, $d = 8$ and $n = 51$ Complete method E.g $n = \frac{900-500}{8}+1$ and $S = \frac{51}{2}\{2\times 900+50\times -8\}$ 35700	B1 M1 A1
		(3)
(ii)	(a) $\frac{11-k}{k-2} = \frac{k-2}{k+4}$ $44-k^2+7k = k^2-4k+4$ $2k^2-11k-40=0$ *	M1 dM1 A1*
		(3)
	(b) $2k^2-11k-40=0 \Rightarrow k = \left(-\frac{5}{2}\right), 8$ Uses either k value and attempts to find both a and r. E.g. with $k=8 \Rightarrow a = 8+4, r = \frac{8-2}{8+4}$ Uses $S_\infty = \frac{a}{1-r} = \frac{12}{1-\frac{1}{2}} = 24$	B1 M1 dM1, A1
		(4)
		Total 13

9 (a)	Substitute $t=4$ in $3\log_2(t+4)-2\log_2(t-2) \Rightarrow 3\log_2 8-2\log_2 2$	M1
	$=3\times3-2\times1=7$ ✓	A1
		(2)
(b)	One correct log law applied. E.g $3\log_2(t+4)=\log_2(t+4)^3$	M1
	Correctly removes logs $\dfrac{(t+4)^3}{(t-2)^2}=2^7$	A1
	$t^3+12t^2+48t+64=128\left(t^2-4t+4\right) \Rightarrow t^3-116t^2+560t-448=0$	A1*
		(3)
(c)	$t^3-116t^2+560t-448=(t-4)\left(t^2+.....\pm112\right)$	M1
	$t^3-116t^2+560t-448=(t-4)\left(t^2-112t+112\right)$	A1
	Solves their $t^2-112t+112$ and finds at least the value of t greater than 2	dM1
	$t=4,\ 56+12\sqrt{21}$	A1
		(4)
		Total 9

10 (a)	$y = \dfrac{9x - x^2}{2\sqrt{x}} = \dfrac{9}{2}x^{\frac{1}{2}} - \dfrac{1}{2}x^{\frac{3}{2}} \Rightarrow \quad \dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{9}{4}x^{-\frac{1}{2}} - \dfrac{3}{4}x^{\frac{1}{2}}$	M1, A1
	Sets $\dfrac{9}{4}x^{-\frac{1}{2}} - \dfrac{3}{4}x^{\frac{1}{2}} = 0 \Rightarrow x = \dfrac{9}{4} \times \dfrac{4}{3} \Rightarrow x = 3$	dM1, A1
		(4)
(b)	$\displaystyle\int \left\{ \frac{9}{2}x^{\frac{1}{2}} - \frac{1}{2}x^{\frac{3}{2}} \right\} \mathrm{d}x = 3x^{\frac{3}{2}} - \frac{1}{5}x^{\frac{5}{2}}$	M1, A1
	Upper limit is 9	B1
	Area $R = \left[3x^{\frac{3}{2}} - \dfrac{1}{5}x^{\frac{5}{2}} \right]_1^9 = \left(3 \times 27 - \dfrac{1}{5} \times 243 \right) - \left(3 \times 1 - \dfrac{1}{5} \times 1 \right)$	dM1
	$= \dfrac{148}{5}$	A1
		(5)
		Total 9